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Python Advanced Assignment - Theory Questions

1. Difference between Iterator and Generator

Iterator:
An iterator is an object that allows sequential traversal of elements using the __iter__() and __next__() methods.

Generator:
A generator is a special type of iterator created using the 'yield' keyword. It generates values one at a time without storing all values in memory.

Example (Iterator):
```
numbers = [10, 20, 30]
iterator = iter(numbers)
next(iterator)
```

Example (Generator):
```
def squares(n):
    for i in range(1, n + 1):
    yield i * i
```

Key Differences:
- Iterator requires implementing __iter__() and __next__().
- Generator uses yield and is easier to implement.
- Generators are more memory efficient for large datasets.

2. Advantages of Using an IDE Debugger over Print Statements

- Execute code line by line.
- Set breakpoints without modifying the source code.
- Inspect variable values during execution.
- View the function call stack.
- Identify logical errors more quickly.
- Modify variable values while debugging.
- More efficient for debugging large applications than using multiple print statements.

3. Difference between Module and Package

Module:
A module is a single Python (.py) file containing functions, classes, or variables.

Example:
utilities.py

Package:
A package is a directory containing multiple Python modules along with an __init__.py file.

Example:
math_package/
    __init__.py
    arithmetic.py
    calculator.py

Key Differences:
• Module is a single Python file.
• Package is a collection of related modules.
• Packages help organize large Python projects.